

UPC INSTRUMENTS PRIVATE LIMITED

Planned Preventive Maintenance (PPM) Schedule

Model: UPC-WA-202

Parameters Monitored: BOD, COD, TSS, pH

1. OBJECTIVE To reduce downtime, prevent unexpected failures, and ensure the long-term accuracy and reliability of the analyser through a structured maintenance program.
2. PPM FREQUENCY & TASK MATRIX

Frequency	Component/Area	Activity	Remarks
weekly	System Auto-Cleaning	Verify 2-hour automated H ₂ SO ₄ dosing cycle executed successfully	Check dosing pump, relay, and solution level
	Display & Readings	Monitor UI screen for abnormal values or errors	Ensure BOD, COD, TSS, pH values are stable
monthly	Sampling Pump	Check suction line, clean inlet strainer	Ensure smooth water intake from ETP
	Dosing Pump & Acid Tank	Check acid level, pump noise, and pipe blockage	Use 5% H ₂ SO ₄ only
	Flow Cell Visual Inspection	Inspect for signs of fouling/cloudiness	Use torchlight if needed
Quarterly	Calibration	Perform 2-point calibration for BOD, COD, TSS, and pH	Use certified standard solutions
	RTU & Wiring	Verify 24V supply, RS485 integrity, CPCB portal communication	Match RTU ID with analyzer serial ID
	Internal Electrical Board	Inspect relays, terminal blocks, and PCB for corrosion or loose connections	Tighten terminals if needed
	UI Touch Display	Check touch sensitivity and software logs	Back up logs manual if needed

	Flow Meter (If installed)	Calibrate manually via Settings → Flow Calibration	Use “Zero” then “Span” based on actual flow
Half yearly	Pump Tubes & Connectors	Replace if hardened, cracked, or leaking	Acid tubing must be chemical-resistant
Annually	Complete System Audit	Full mechanical and electrical inspection + calibration + cleaning	Generate service report
	UV Lamp Health Check	Check xenon lamp intensity and replacement need	By passing different sample

3. SPARE PARTS RECOMMENDATION (ANNUAL STOCK)

- pH Electrode – 1 No.
- Acid Dosing Pump Tube – 2 sets
- UV Lamp (Xenon) – 1 No.
- Relay Board (spare) – 1 No.
- 5% H₂SO₄ (Stock) – 10 Ltr minimum
- Sensor Cleaning Brushes – 1 set

4. RECORD KEEPING Each maintenance activity should be logged in the PPM Register / Maintenance Logbook with:

- Date & Time
- Component Maintained
- Activity Type
- Performed By (Technician Name)
- Observations/Issues
- Signature

5. NOTES & SAFETY INSTRUCTIONS

- Always use PPE (Gloves, Safety Glasses, Lab Coat) when handling H₂SO₄ or opening flow cell.
- Disconnect power supply during manual sensor maintenance.
- Use only OEM-approved accessories and calibration solutions.
- In case of persistent drift or erratic readings, contact QESPL Technical Support.

6. SUPPORT & SERVICE CONTACT UPC INSTRUMENTS PRIVATE LIMITED

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